TEAM CODE: [INSERT TEAM CODE]

(Submission strictly adheres to anonymity guidelines - no participant or institution names)

CodeRx: An Intelligent Longitudinal Health Record Platform and Differential Clinical Decision Support System

Outpatient healthcare delivery is severely constrained by fragmented medical records, hurried consultations, and clinician cognitive overload. Patients routinely present with disjointed laboratory PDFs, physical prescriptions, and unstructured diagnostic histories scattered across disparate healthcare providers. Consequently, physicians spend disproportionate consultation time manually sifting through historical documents rather than delivering focused clinical care. Crucially, subtle yet dangerous biomarker shifts, medication discrepancies, chronic disease progressions, and adverse drug-drug interactions frequently go undetected during brief outpatient encounters, jeopardizing long-term treatment outcomes.

To resolve these systemic inefficiencies, this project presents CodeRx, a unified, desktop-first healthcare management platform and clinical decision support system. CodeRx transforms unstructured diagnostic scans, lab reports, and prescriptions into a cohesive, chronological medical timeline. Unlike existing electronic health record systems that function primarily as passive document repositories, CodeRx introduces an autonomous 'What Changed?' differential analytics engine. Leveraging constrained multimodal artificial intelligence, the platform extracts discrete biochemical parameters across longitudinal encounters, dynamically categorizing biomarker trajectories into improved, stable, or acute review states. It automatically generates a sixty-second 'Doctor Brief' outlining critical clinical changes, stopped medications, and targeted consultation focal points prior to patient examination.

The platform bridges clinical decision-making and patient empowerment through integrated teleconsultations, digital prescription generation with deprescribing safety checks, and structured pre-visit diagnostic checklists. Designed specifically for outpatient clinicians, multimorbid chronic care patients, and ambulatory clinics, CodeRx operates under rigorous clinical safety boundaries with zero autonomous diagnostic extrapolation and verifiable raw-data auditability. By converting fragmented documentation into actionable longitudinal intelligence, CodeRx minimizes diagnostic oversight, curbs adverse polypharmacy events, accelerates clinical workflows, and fosters standardized, patient-centric healthcare delivery.

Keywords: Clinical Decision Support Systems, Longitudinal Health Records, Healthcare Informatics, Differential Analytics, Polypharmacy Management, Patient Safety.